

strategy for the second player. In this second case, instead of eating just the cookie in the bottom right corner, the first player could have made the same move that the second player made as the first move of a winning strategy (and then continued to follow that winning strategy). This would guarantee a win for the first player.

Q) How can the first player apply the would-be strategy of the second player since the state of game is different, i.e. the second player would have applied the strategy with the rightmost cookie removed at the bottom?

Note: No one has been able to describe a winning strategy for Chomp that applies for all rectangular grids by describing the moves that the first player should follow.

1.8.4 Uniqueness Proofs

Some theorems assert the existence of a unique element with a particular property. In other words, these theorems assert that there is exactly one element with this property. To prove a statement of this type we need to show that an element with this property exists and that no other element has this property.

The two parts of a uniqueness proof are: ① Existence: We show that an element x with the desired property exists. ② Uniqueness: We show that if both x and y have the desired property, then $x=y$.

Example 13: Show that if a and b are real numbers and $a \neq 0$, then there is a unique real number r such that $ar+b=0$.

Solution: Existence: $ar+b=0 \Rightarrow r = -(\frac{b}{a})$
 $ar+b = a(-\frac{b}{a})+b = (-b)+b=0$

$\therefore$ A real number r exists such that $ar+b=0$.